

MSc THESIS • JULY 2025

The Sustainability Paradox

How AI is Revolutionizing Sustainability Reporting While Creating New Challenges

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Why this topic matters

01

CSRD raises the bar

The EU Corporate Sustainability Reporting Directive demands more detailed ESG disclosure.

02

Reporting becomes data-heavy

More indicators, evidence and workflows increase reporting complexity.

03

AI-based SaaS scales the work

Companies adopt automation—but AI introduces a footprint of its own.

The research question

How does AI-enabled sustainability reporting create value—and new sustainability risks?

1

Reporting efficiency

2

Accuracy & reliability

3

Compliance outcomes

4

Environmental impact

What sustainability reporting is



A recurring process—not a one-off document.

Three kinds of tools compared

Manual / spreadsheet

Human-led collection and consolidation

Flexible • labour-intensive

Rules-based software

Configured workflows and validations

Consistent • configuration-dependent

AI-enabled SaaS

Machine-assisted extraction, analysis and drafting

Scalable • model-dependent

What AI adds, and what it costs

WHAT AI ADDS

- Faster extraction and consolidation
- Automated classification and drafting
- Pattern detection across large datasets
- More scalable reporting workflows

WHAT AI COSTS

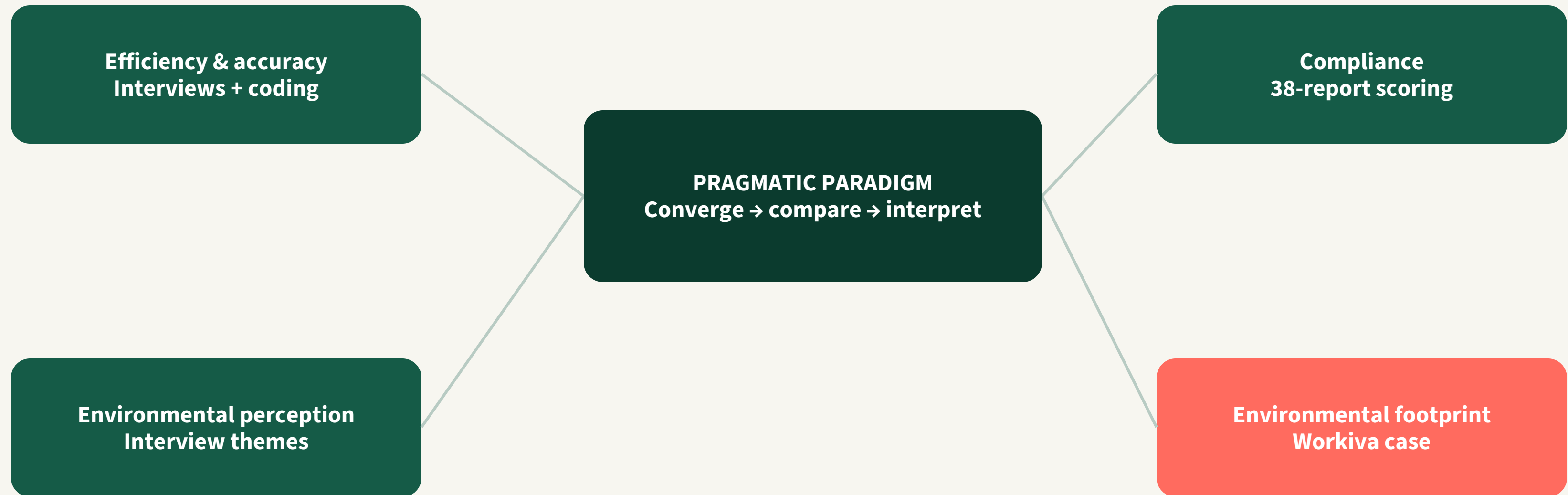
- Energy and infrastructure demand
- Opaque models and uncertain provenance
- Bias, privacy and accountability risks
- Risk of automating weak source data

Theory lens: paradox theory and systems thinking



Paradox theory (Hahn et al.) argues that organisations should accept and manage persistent tensions—not try to eliminate them.

Research design: one question, four methods



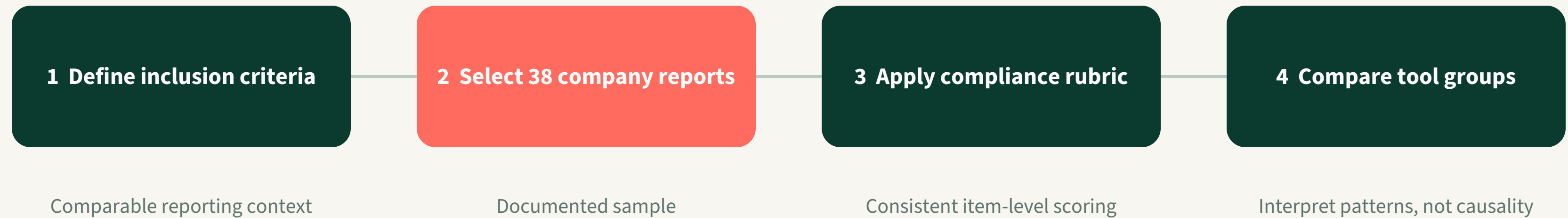
Creswell and Plano Clark, 2018

Qualitative method: interviews and coding



Counts are retained in the coded thesis dataset; interpretation moves from participant language to cross-case dimensions.

Quantitative method: scoring compliance



The score tests observable compliance in published reports—not hidden organisational capability.

Environmental impact method: the Workiva case



Why Workiva? A bounded, transparent case enables a defensible attribution exercise while keeping assumptions visible.

Finding 1: Efficiency, the clearest win

FASTER

Collection and consolidation

LESS MANUAL

Repetitive reporting work

MORE SCALABLE

High-volume disclosure workflows

Finding 2: Accuracy, faster but not more accurate

WHAT IMPROVED

01 Consistency checks

02 Anomaly detection

03 Traceable workflows

WHAT DID NOT

01 Weak source data persists

02 Human review remains essential

03 Confidence can exceed evidence

Finding 3: Compliance, better only where data is structured

STRUCTURED DATA
Clearer compliance advantage

AI-enabled workflows perform best when inputs, rules and evidence are standardised.

UNSTRUCTURED / JUDGMENT-HEAVY
Advantage weakens

Interpretation, materiality and narrative quality still depend on human judgment.

Technology amplifies data readiness; it does not substitute for it.

Finding 4a: Environmental impact, what practitioners said

01 Low visibility

Footprints are rarely visible in day-to-day use.

02 Efficiency optimism

Automation is assumed to offset some burden.

03 Materiality uncertainty

Practitioners struggle to judge significance.

04 Growing governance need

Measurement and vendor disclosure must improve.

Finding 4b: Environmental impact, the Workiva numbers

2024

Reporting year used

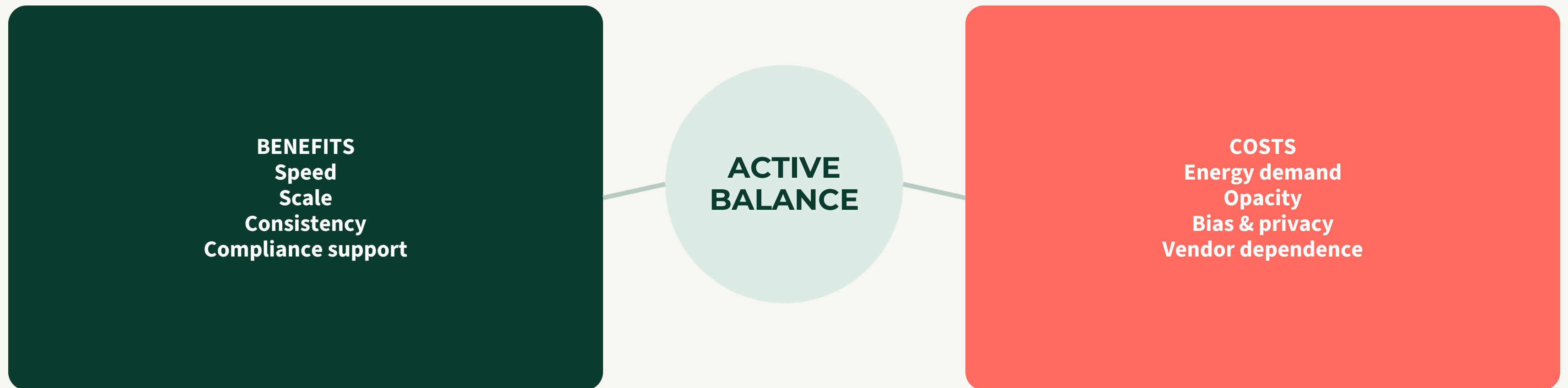
20%

Share attributed to sustainability
software

ATTRIBUTED

Case-based allocation—not a direct
meter reading

The paradox in one picture



The goal is not frictionless AI. It is governed AI that keeps both sides visible.

What should change

01 Govern the use case

Match automation to risk and materiality.

02 Strengthen source data

Standardise inputs before adding intelligence.

03 Measure the footprint

Require vendor transparency and attribution.

04 Keep humans accountable

Preserve review, challenge and sign-off.

Limitations and future research

LIMITATIONS

- Bounded interview and report samples
- Published reports reveal outcomes, not internal process
- Workiva attribution depends on assumptions
- Rapidly changing AI capabilities

FUTURE RESEARCH

- Replicate across sectors and jurisdictions
- Track reporting quality longitudinally
- Compare model and vendor footprints
- Test governance practices in use

CONCLUSION

**AI can improve sustainability reporting.
It cannot make the underlying tension disappear.**

The responsible path is to pair capability with data discipline, environmental accounting and human accountability.